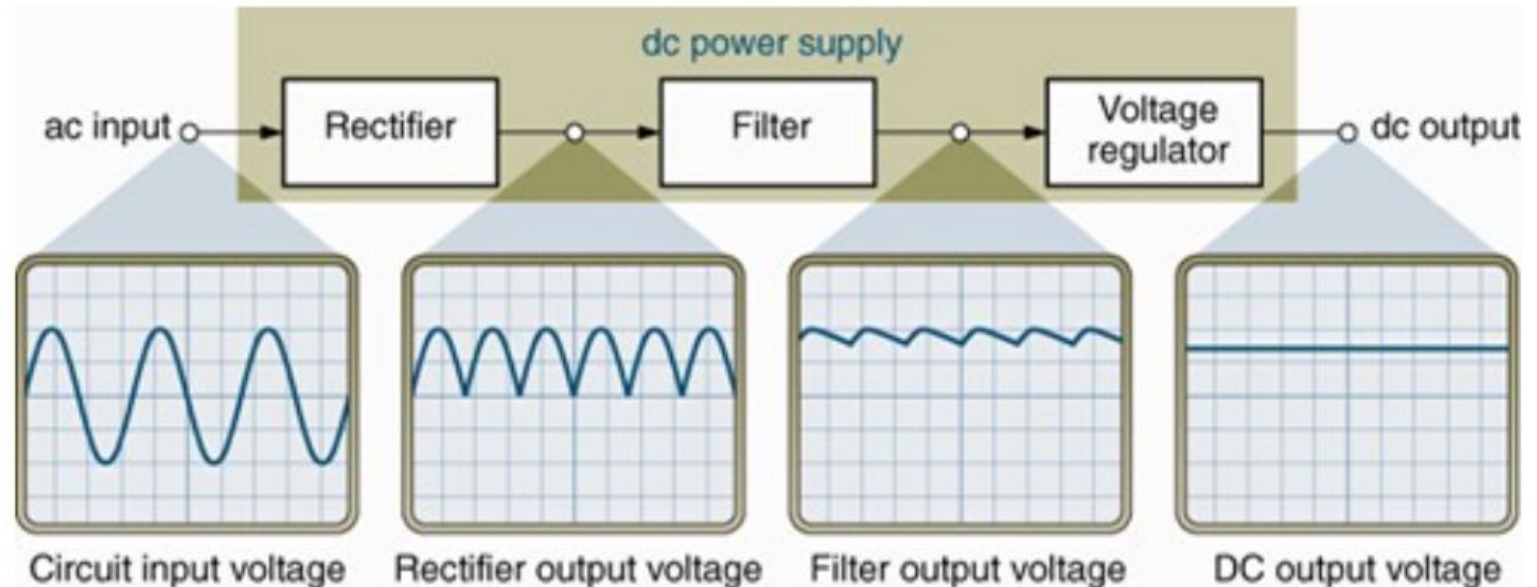


Chapter 3

DC Power Supply

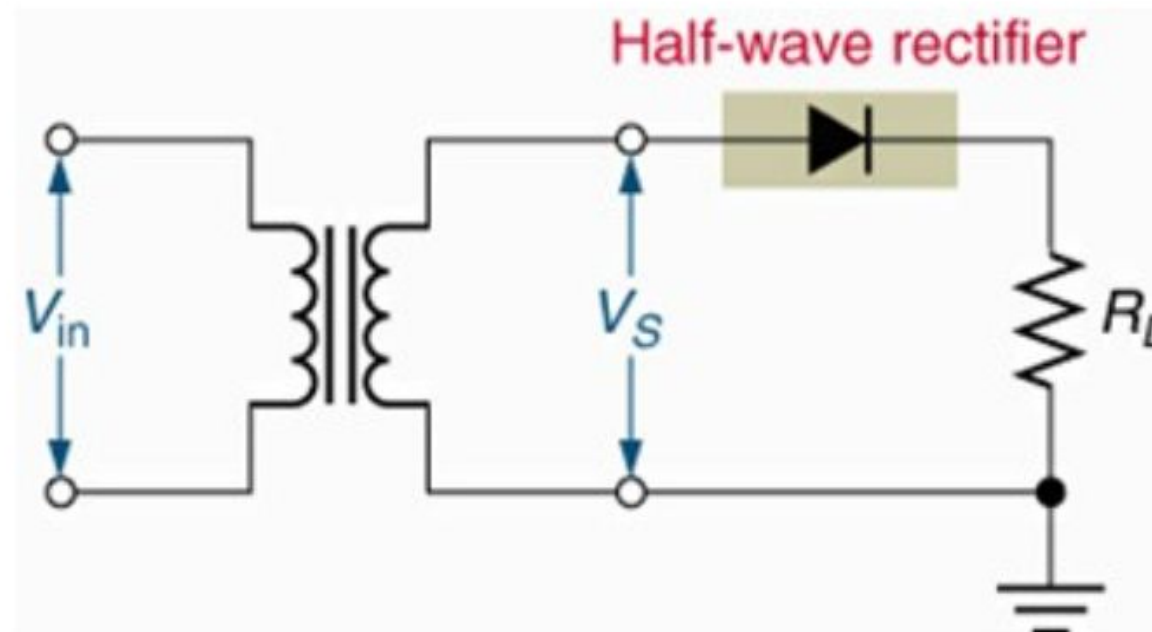
Power Supply

- A group of circuits used to convert ac to dc.
 - **Rectifier** – Converts ac to pulsating dc.
 - **Filter** – Reduces variations in the rectifier output.
 - **Voltage regulator** – Maintains a constant dc output voltage.



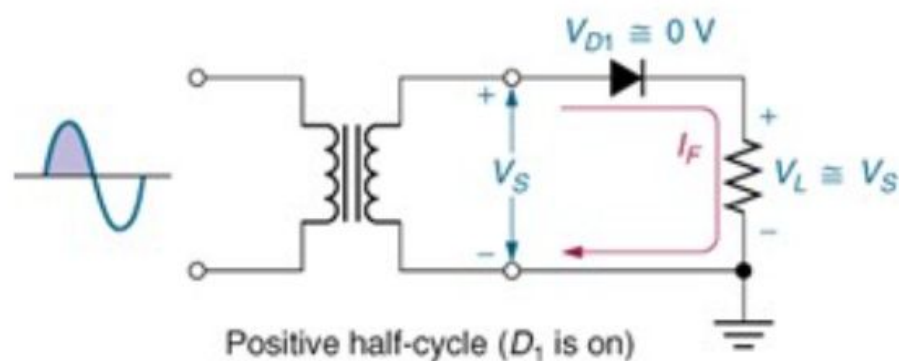
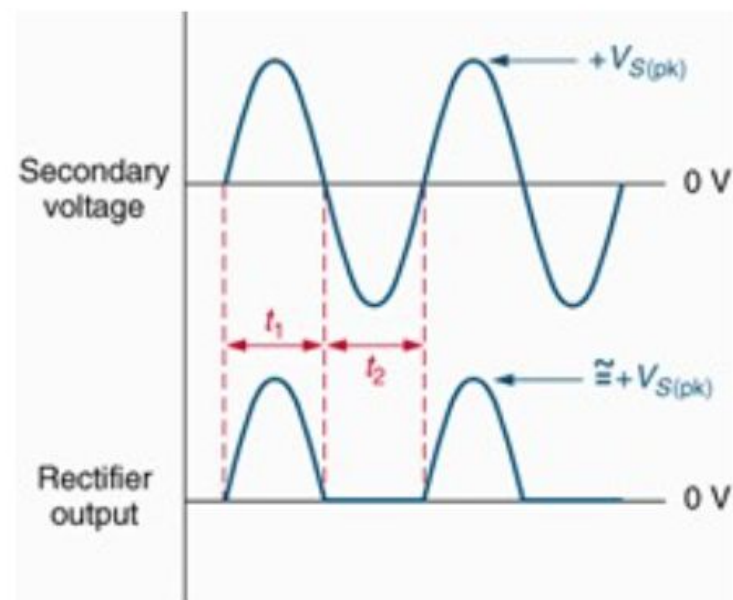
Half-wave Rectifiers

- **Half-wave rectifier** – A diode placed in series between a transformer (or ac line input) and its load.

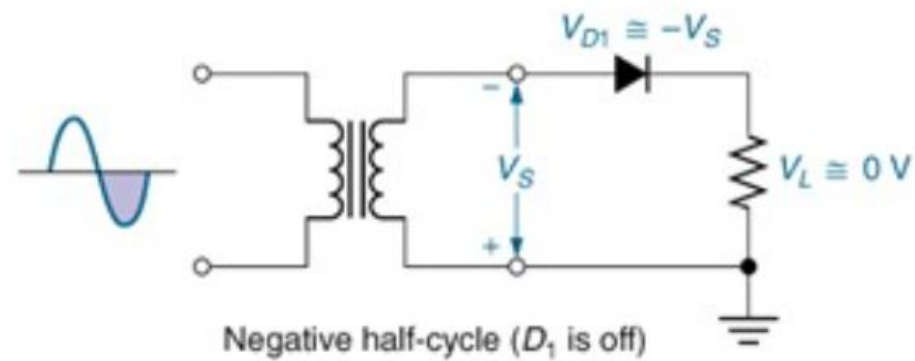


Positive Half-wave Rectifiers

This circuit converts an ac input to a series of positive pulses.



(a)



(b)



Average Load Voltage and Current

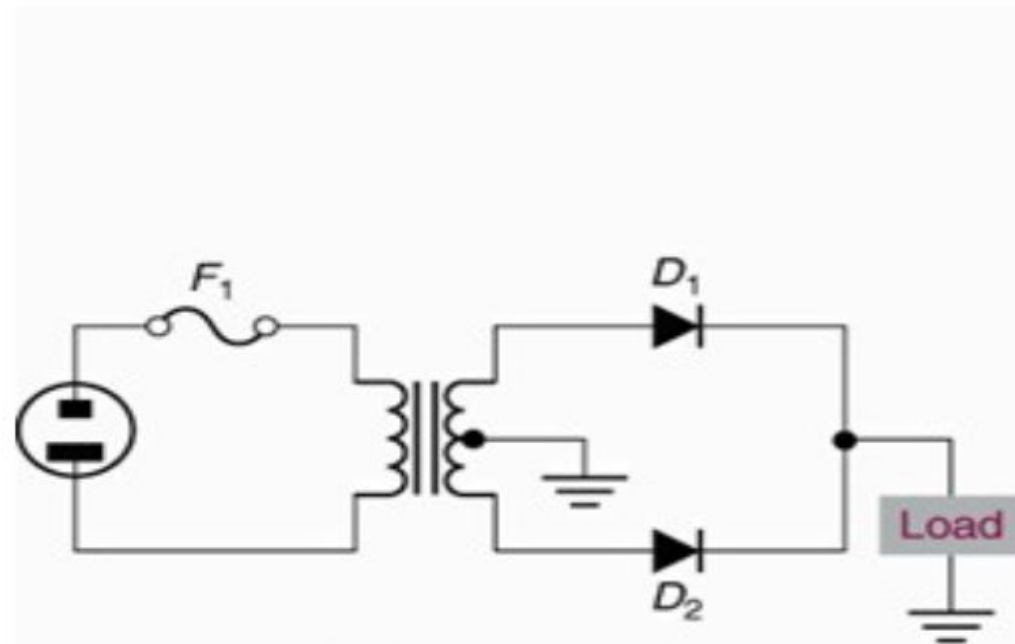
- **Average voltage (V_{ave})** – The dc equivalent of a voltage waveform.
- **Average current (I_{ave})** – The dc equivalent of a current waveform.

For the output from a *half-wave* rectifier:

$$V_{ave} = \frac{V_{pk}}{\pi}$$

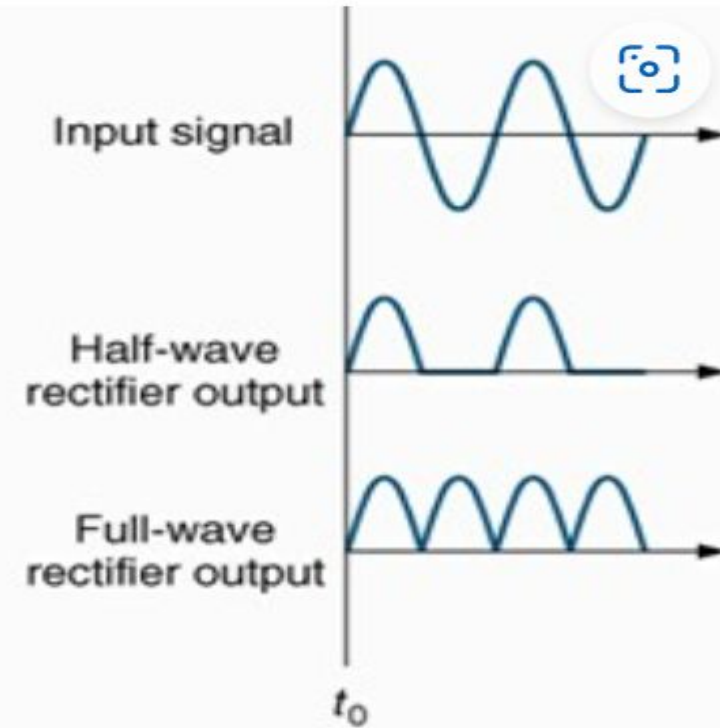
$$I_{ave} = \frac{I_{pk}}{\pi}$$

Full wave Center Tapped Rectifier



A full-wave rectifier

(a)

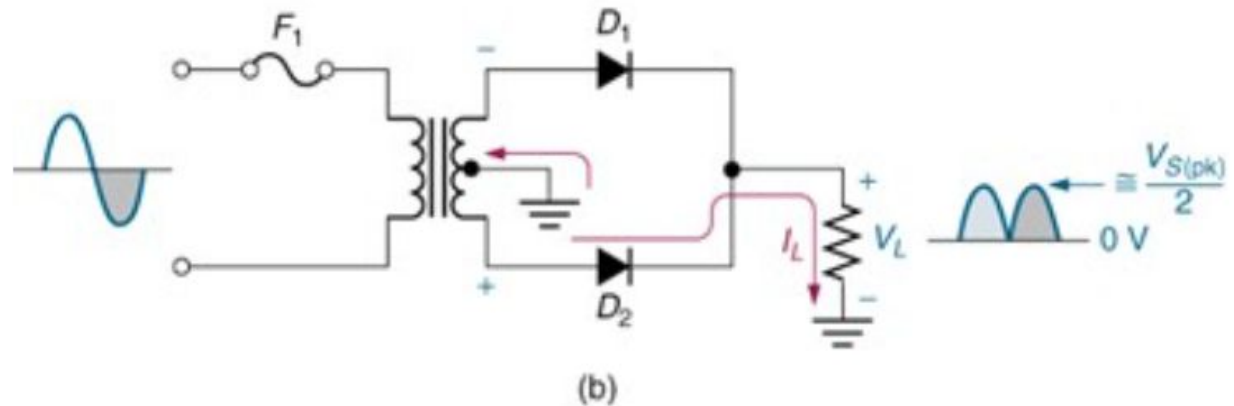
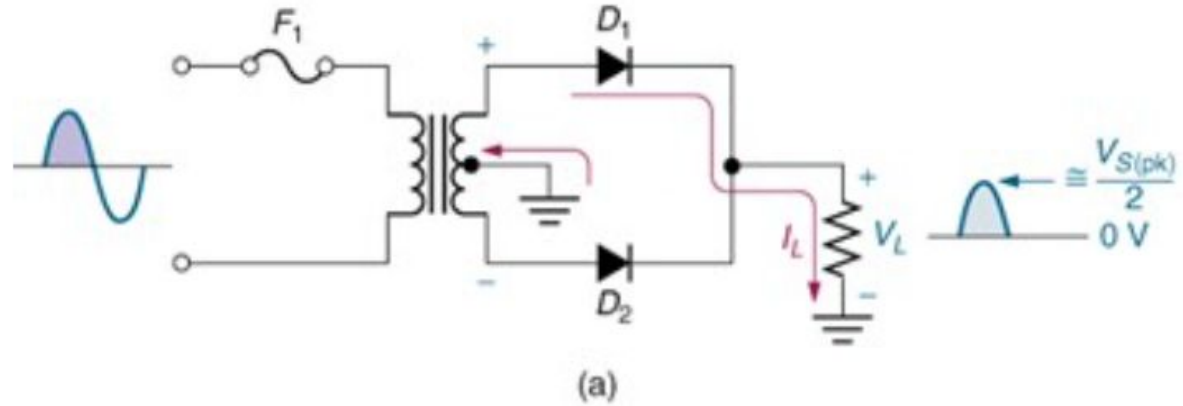


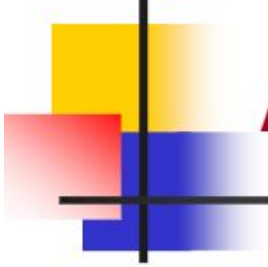
Typical rectifier waveforms

(b)

Full-wave Rectifier Operation

- Diodes conduct during alternate half cycles of the input signal.
- $V_{L(pk)}$ is approximately half the value of $V_{S(pk)}$.
- The circuit produces two positive half-cycles for each input cycle.





Average Load Voltage and Current

- Average voltage (V_{ave}) – The dc equivalent of a voltage waveform.
- Average current (I_{ave}) – The dc equivalent of a current waveform.

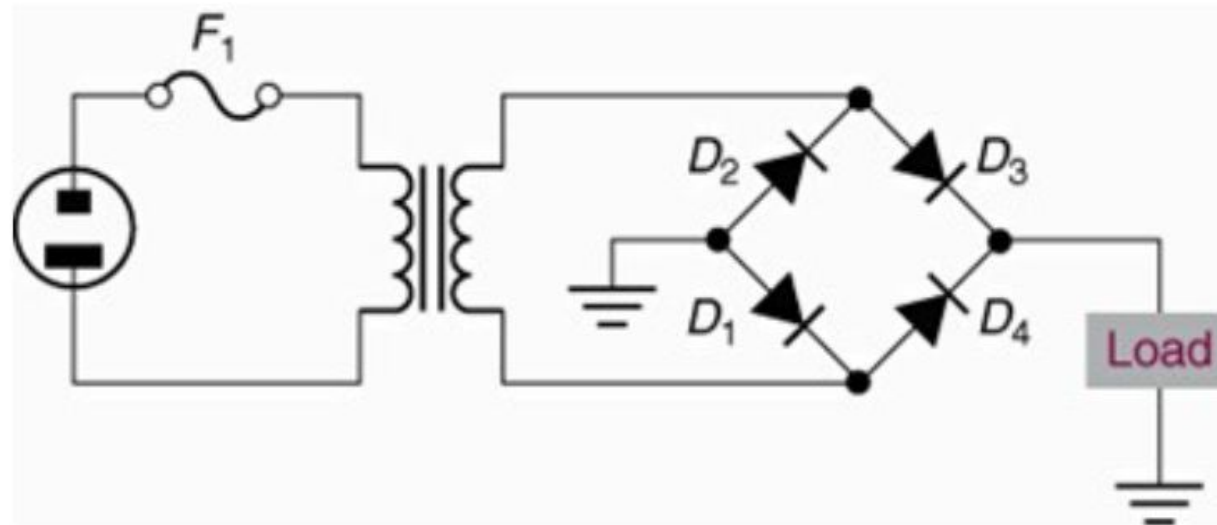
For the output from a full-wave rectifier:

$$V_{ave} = \frac{2V_{pk}}{\pi}$$

$$I_{ave} = \frac{2I_{pk}}{\pi}$$

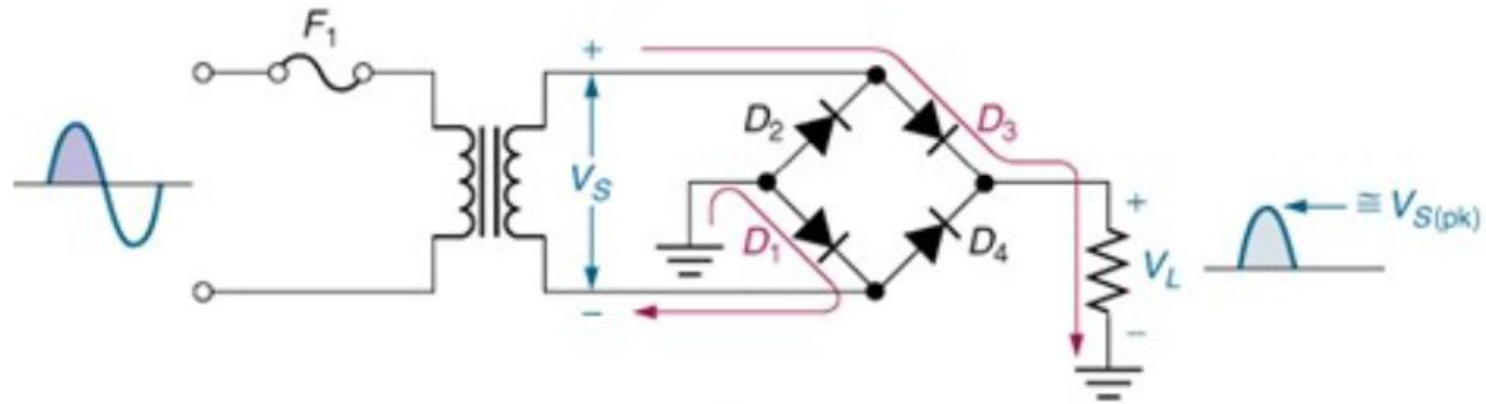
Full-Wave Bridge Rectifiers

- The most commonly used because:
 - It does not require the use of a center-tapped transformer.
 - It can be coupled directly to the ac power line.
 - It produces a higher dc output than a comparable full-wave center-tapped rectifier.

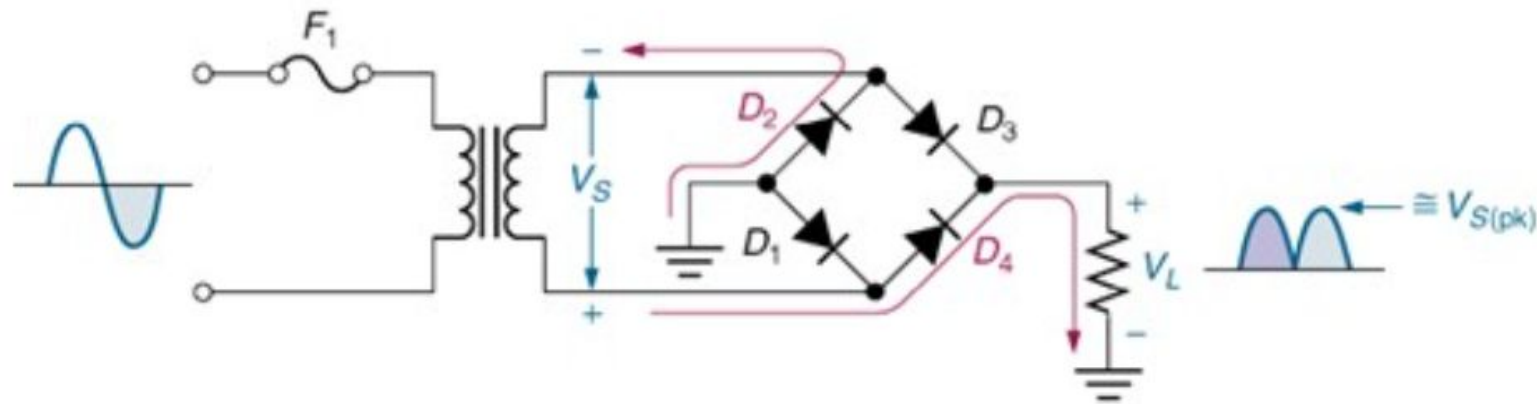


Bridge Rectifier Operation

Conduction alternates between two diode pairs.

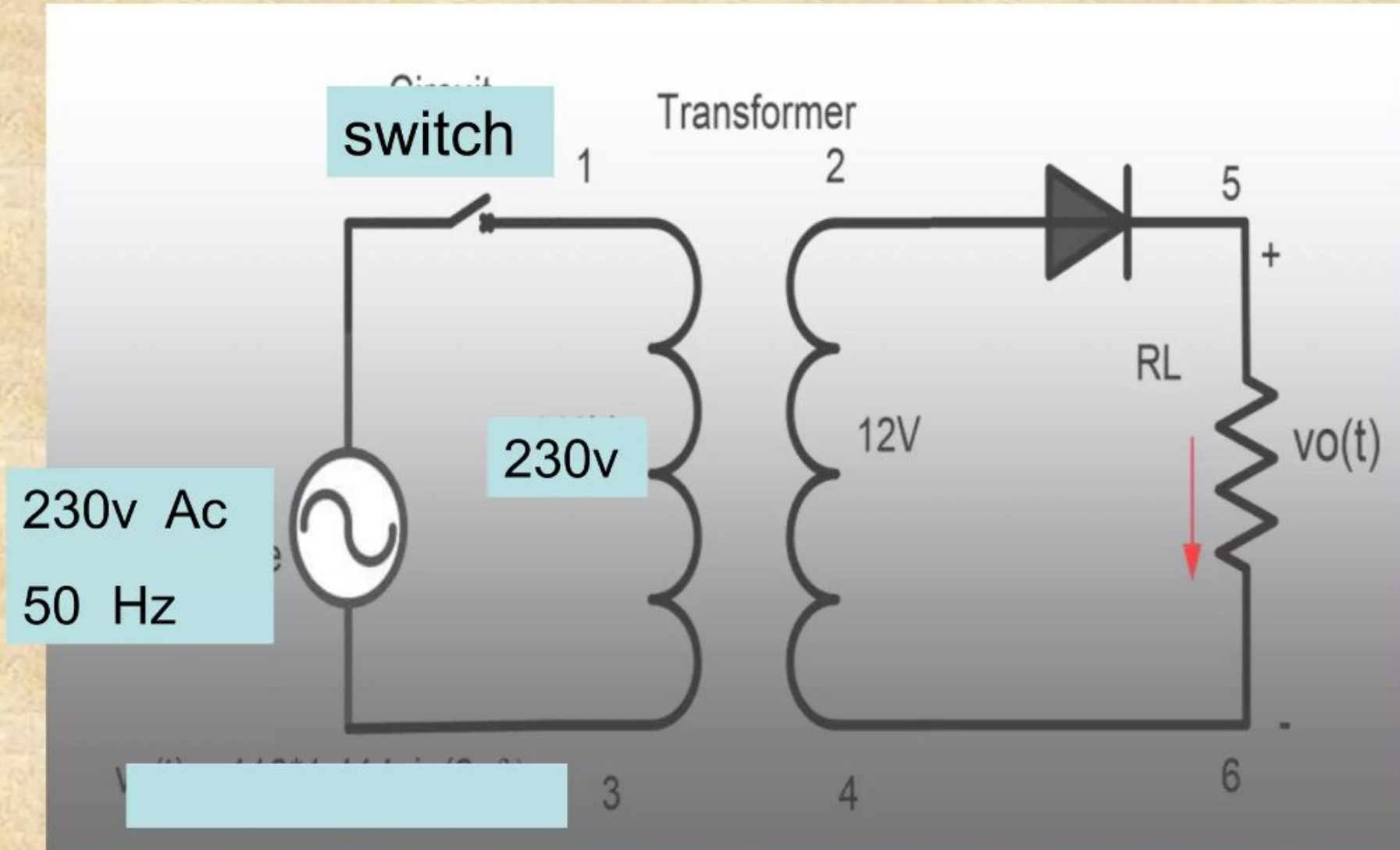


(a)



(b)

The Half-wave Rectifier Circuit (without filter)



Transformer secondary voltage

$$V_{in} = V_m \sin \omega t$$

V out Calculation

$$v(t) = \begin{cases} V_m \sin \omega t & \text{for } t \leq \frac{T}{2} \\ 0 & \text{for } \frac{T}{2} < t < T \end{cases}$$

Compute the average voltage (dc voltage) V_{avg} or V_{dc} for one complete cycle

$$V_{dc} = V_{avg}$$

For a half wave rectifier, $V_{rms} = V_m/2$
and $V_{dc} = V_m/\pi$

$$V_{dc} = \frac{1}{T} \int_0^T v(t) dt$$

$$= \frac{1}{2\pi} \left[\int_0^{\pi} V_m \sin \omega t dt + \int_{\pi}^{2\pi} 0 dt \right]$$

$$= \frac{V_m}{2\pi} [-\cos \omega t]_0^{\pi} = \frac{V_m}{2\pi} [-\cos(\pi) - (-\cos 0)]$$

$$= \frac{V_m}{2\pi} [-(-1) - (-1)] = \frac{V_m}{\pi} = 0.318 V_m$$

V_{rms} at the Load Resistance

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt} \quad v(t) = \begin{cases} V_m \sin \omega t & \text{for } t \leq \frac{T}{2} \\ 0 & \text{for } \frac{T}{2} < t < T \end{cases}$$

$$= \left[\frac{1}{2\pi} \left[\int_0^{T/2} V_m^2 \sin^2 \omega t d(\omega t) + \int_{T/2}^T 0 d(\omega t) \right] \right]^{1/2}$$

$$= \left[\frac{V_m^2}{2\pi} \int_0^{T/2} \sin^2 \omega t d(\omega t) \right]^{1/2}$$

$$\sin^2 \omega t = \frac{1}{2} (1 - \cos 2\omega t), \quad \omega T = 2\pi, \quad \theta = \omega t$$

$$V_{rms} = \left[\frac{V_m^2}{4\pi} \int_0^\pi (1 - \cos 2\theta) d\theta \right]^{1/2}$$

$$= \left[\frac{V_m^2}{4\pi} \left(\theta - \frac{1}{2} \sin 2\theta \right) \Big|_0^\pi \right]^{1/2}$$

$$= \left[\frac{V_m^2}{4\pi} \left(\pi - \frac{1}{2} \sin(2\pi) - 0 + \frac{1}{2} \sin 2(0) \right) \right]^{1/2}$$

$$= \left[\frac{V_m^2}{4\pi} (\pi - 0 - 0 + 0) \right]^{1/2}$$

$$= \frac{V_m}{2}$$

$$= V_m / 2$$

Ripple Factor = RMS value of the AC component/DC value of the component

$$\begin{aligned} \text{Ripple} &= \frac{V_{rms}}{V_{dc}} \\ &= \frac{\sqrt{V_{rms}^2 - V_{dc}^2}}{V_{dc}} \\ &= \sqrt{\left[\frac{V_{rms}}{V_{dc}} \right]^2 - 1} \\ &= \sqrt{\left[\frac{V_m / 2}{V_m / \pi} \right]^2 - 1} \\ &= \sqrt{\left[\frac{\pi}{2} \right]^2 - 1} \\ &= 1.2114 \end{aligned}$$

Efficiency

$$\eta = (\text{dc output power} / \text{ac input power}) \times 100\%$$

$$\eta = p_{dc} / p_{ac}$$

$$\text{Where } p_{dc} = V_{dc}^2 / R_L \text{ and } p_{ac} = V_{rms}^2 / R_L$$

$$\eta = \frac{V_{dc}^2 / R_L}{V_{rms}^2 / R_L} = \frac{(V_m / \pi)^2}{(V_m / 2)^2} = \frac{4}{\pi^2} = 0.406$$

or

$$\eta = 40.6\%$$

Form Factor

Form Factor = rms value / average value

$$= (V_{m/2}) / (V_{m/\pi}) = \pi/2 = 1.11$$

Peak Factor

Peak factor = Peak value / rms value = $V_m / (V_m/2) = 2$

Comparison with centre tapped F.W.R

Centre tapped FWR

- No. of diodes are two
- PIV is $2V_m$
- Bigger secondary of transformer is required
- Preferred for low voltage and high current circuits

Bridge FWR

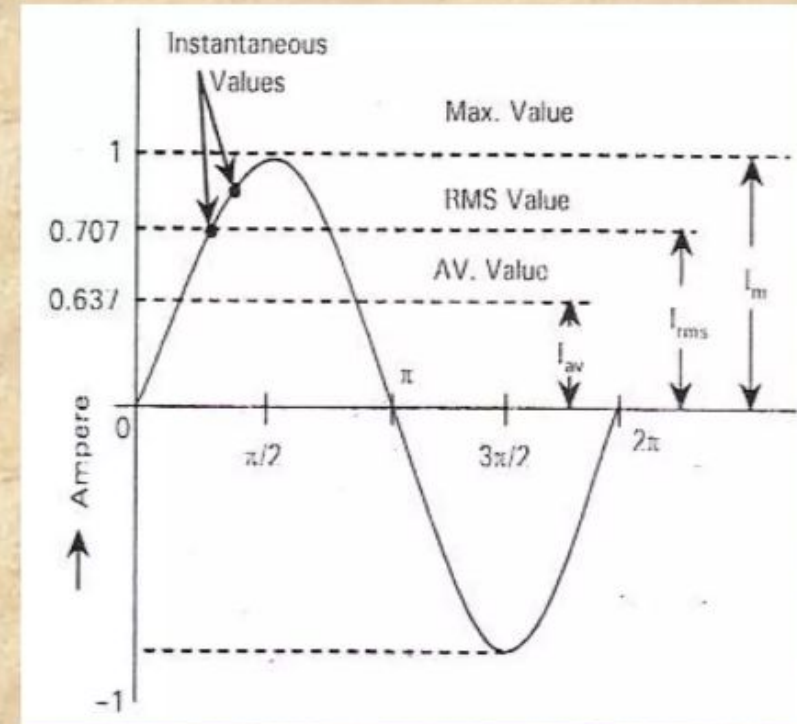
- No. of diodes are four
- PIV is V_m
- Secondary of transformer requires half windings for the same voltage
- Preferred for low voltage and high current circuits

RMS Value of FWR

$$i_L = I_m \sin \omega t$$

RMS value of the voltage at the load resistance is

$$\begin{aligned} I_{rms} &= \sqrt{\frac{1}{\pi} \int_0^{\pi} i_L^2 d(\omega t)} = \sqrt{\frac{1}{\pi} \int_0^{\pi} I_m^2 \sin^2 \omega t d(\omega t)} \\ &= \sqrt{\left(\frac{I_m^2}{\pi} \right) \int_0^{\pi} \left(\frac{1 - \cos 2\omega t}{2} \right) d(\omega t)} \\ &= \sqrt{\left(\frac{I_m^2}{\pi} \right) \left[\frac{\omega t}{2} - \frac{\sin 2\omega t}{4} \right]_0^{\pi}} \\ &= \sqrt{\frac{I_m^2}{\pi} \cdot \frac{\pi}{2}} = \frac{I_m}{\sqrt{2}} \end{aligned}$$



Average Value of FWR

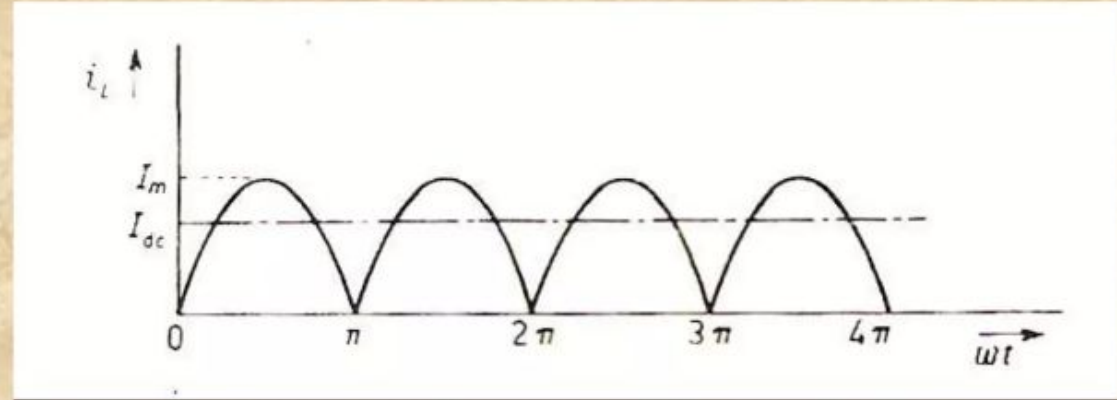
$$I_{dc} = \frac{1}{\pi} \int_0^{\pi} i_L d(\omega t)$$

$$= \frac{1}{\pi} \int_0^{\pi} I_m \sin \omega t d(\omega t)$$

$$= \frac{I_m}{\pi} \left| -\cos \omega t \right|_0^{\pi}$$

$$= \frac{I_m}{\pi} (+1 + 1)$$

$$= \frac{2I_m}{\pi}$$



Ripple factor of FWR

$$= \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1}$$

For a full wave rectifier, $V_{rms} = V_m/\sqrt{2}$, $V_{dc} = 2V_m/\pi$

$$= \sqrt{\left(\frac{\frac{I_m}{\sqrt{2}}}{\frac{2I_m}{\pi}}\right)^2 - 1}$$

$$= \sqrt{\left(\frac{\pi}{2\sqrt{2}}\right)^2 - 1}$$

$$= 0.482$$

Efficiency of FWR

Efficiency of the rectifier

$$\eta = \frac{\text{dc output power}}{\text{ac input power}}$$

The dc output power of FWR

$$\begin{aligned} P_{dc} &= I_{dc}^2 R_L \\ &= \left(\frac{2I_m}{\pi} \right)^2 R_L \end{aligned}$$

The ac input power of FWR

$$\begin{aligned} P_{ac} &= I_{rms}^2 (r_d + R_L) \\ &= \left(\frac{I_m}{\sqrt{2}} \right)^2 (r_d + R_L) \end{aligned}$$

Efficiency of the FWR

$$\eta = \frac{\text{dc output power}}{\text{ac input power}}$$

$$\eta = \frac{\left(\frac{2I_m}{\pi}\right)^2 R_L}{\left(\frac{I_m}{\sqrt{2}}\right)^2 (r_d + R_L)} = \frac{8}{\pi^2} \frac{1}{\left(1 + \frac{r_d}{R_L}\right)}$$

$$= \frac{0.812}{\left(1 + \frac{r_d}{R_L}\right)}$$

$$\% \eta = \frac{81.2}{\left(1 + \frac{r_d}{R_L}\right)}$$

*For ideal diode $r_d = 0$
then, efficiency $\eta = 81.2$*

Number of Diodes	1	2	4
D.C Current	I_m / π	$2 I_m / \pi$	$2 I_m / \pi$
Transformer Necessary	No	Yes	No
Max Value of Current	$V_m / (r_f + R_L)$	$V_m / (r_f + R_L)$	$V_m / (2r_f + R_L)$
Ripple Factor	1.21	0.482	0.482
O/P Frequency	f_{in}	$2 f_{in}$	$2 f_{in}$
Max Efficiency	40.6%	81.2%	81.2%
Peak Inverse Voltage	V_m	$2 V_m$	$2 V_m$

Series Shunt and biased clipper circuits

What is a Clipping Circuit?

Clipping circuit is a wave-shaping circuit, and is used to either remove or clip a portion of the applied wave in order to control the shape of the output waveform.

One of the most basic clipping circuit is the half-wave rectifier. A half-wave rectifier clips either the negative half cycle or the positive half cycle of an alternating waveform, and allows to pass only one half cycle.

Clipping circuits are also referred to as voltage limiters, amplitude selectors, or slicers.

Clippers are used to eliminate amplitude noise or to fabricate new waveforms from an existing signal.

Classification Of Clippers

Clipping circuit consists of non-linear and linear devices. The non-linear devices generally used for clipping are diodes and transistors.

According to non-linear devices used, clippers may be classified as:

- *Diode Clippers*
- *Transistor Clippers*

According to biasing, the clippers may be classified as:

- *Unbiased Clippers*
- *Biased Clippers*

According to configuration used the clippers may be:

- *Series diode clippers*
- *Parallel or shunt diode clippers*
- *A series combination of diode, resistor and reference supply*
- *Multi-diode clippers consisting of several diodes, resistors and reference voltages*

According to level of clipping the clippers may be:

- *Positive clippers*
- *Negative clippers*
- *Biased clippers*
- *Combination clippers*

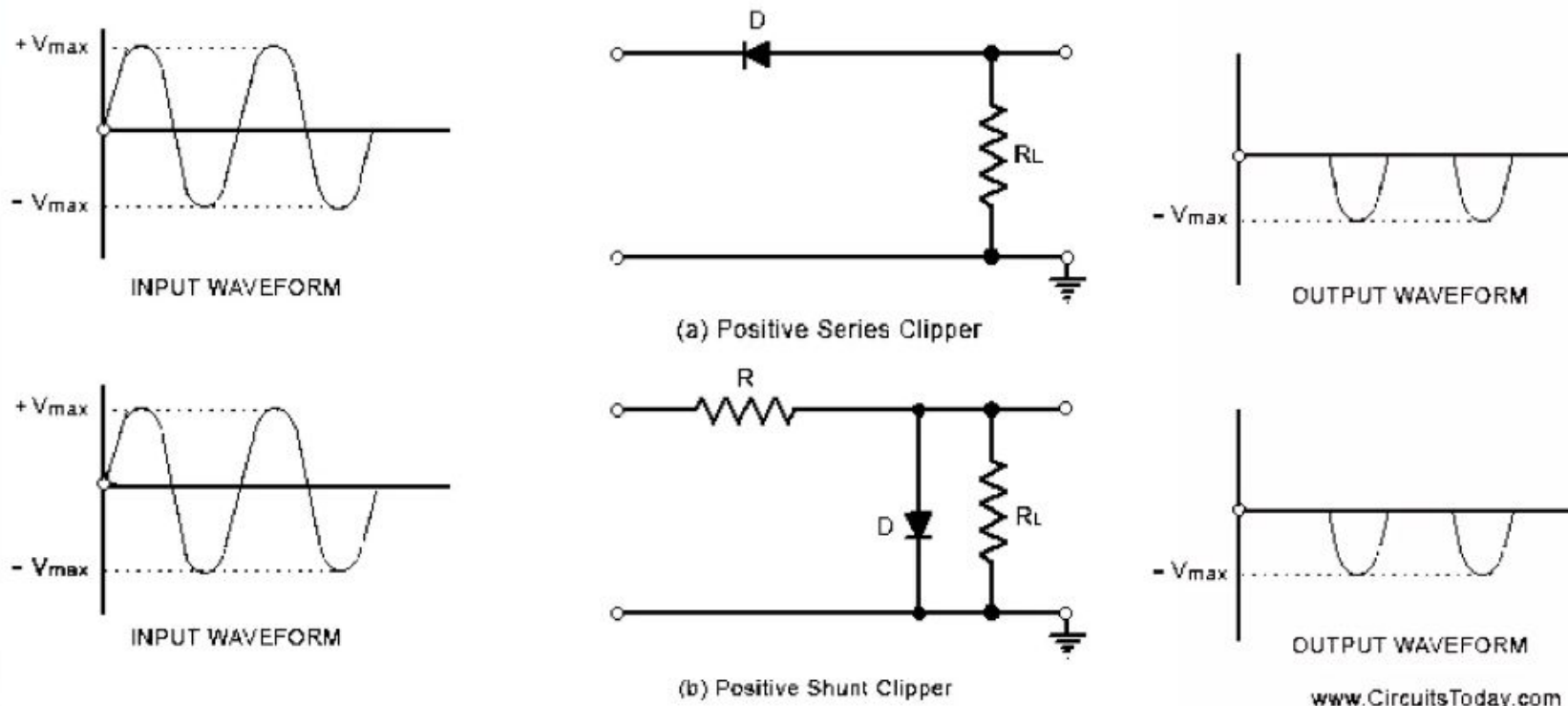
There are two general categories of clippers: series and parallel (or shunt). The series configuration is defined as one where diode is in series with the load, while the shunt clipper has the diode in a branch parallel to the load.

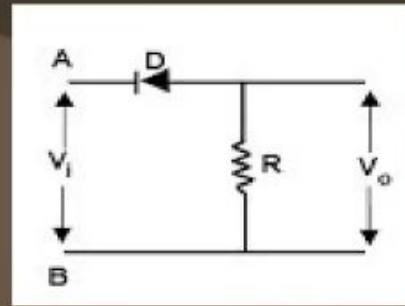
1. Positive Clipper and Negative Clipper

Positive Diode Clipper:

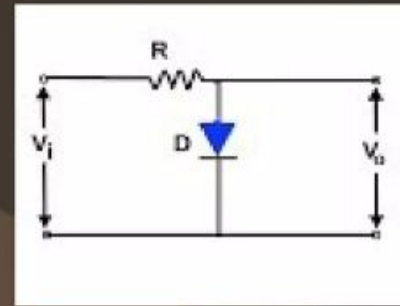
In a positive clipper, the positive half cycles of the input voltage will be removed. The circuit arrangements for a positive clipper is illustrated in the figure given below.

Positive Series Clipper and Positive Shunt Clipper





(a)



(b)

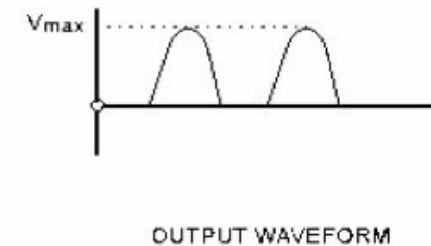
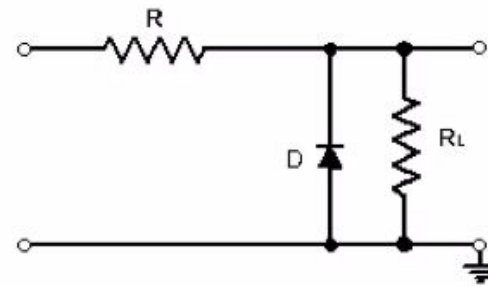
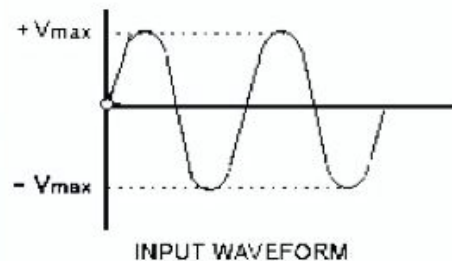
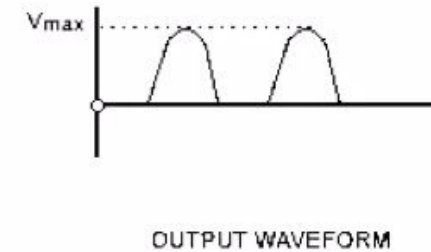
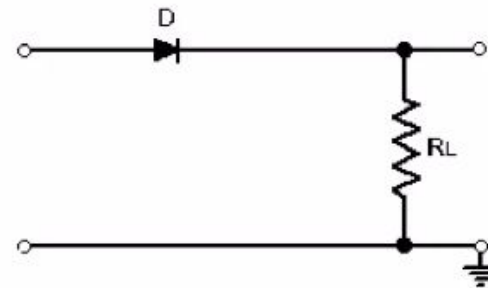
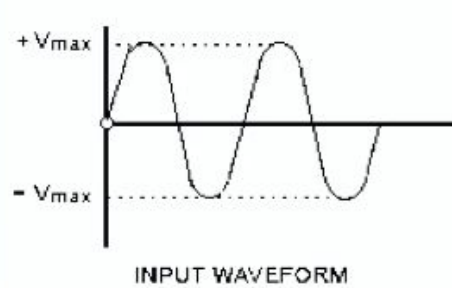
As shown in the Figure(a), the diode is kept in series with the load. During the positive half cycle of the input waveform, the diode 'D' is reverse biased, which maintains the output voltage at 0 Volts. Thus causes the positive half cycle to be clipped off. During the negative half cycle of the input, the diode is forward biased and so the negative half cycle appears across the output.

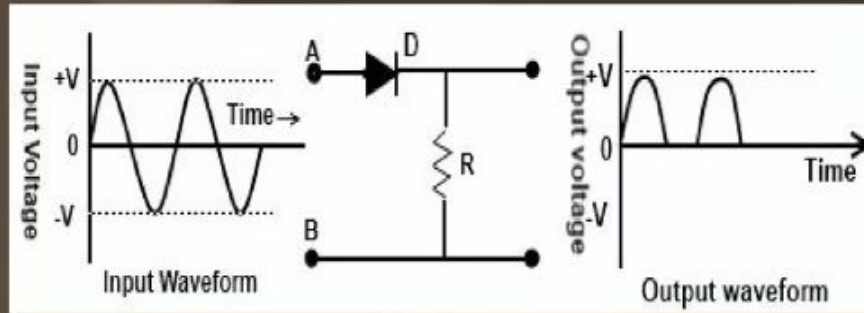
In Figure (b), the diode is kept in parallel with the load. This is the diagram of a positive shunt clipper circuit. During the positive half cycle, the diode 'D' is forward biased and the diode acts as a closed switch. This causes the diode to conduct heavily. This causes the voltage drop across the diode or across the load resistance R_L to be zero. Thus output voltage during the positive half cycles is zero, as shown in the output waveform. During the negative half cycles of the input signal voltage, the diode D is reverse biased and behaves as an open switch. Consequently the entire input voltage appears across the diode or across the load resistance R_L if R is much smaller than R_L .
 Actually the circuit behaves as a voltage divider with an output voltage of $[R_L / R + R_L]$
 $V_{max} = -V_{max}$ when $R_L \gg R$

Negative Diode Clipper:

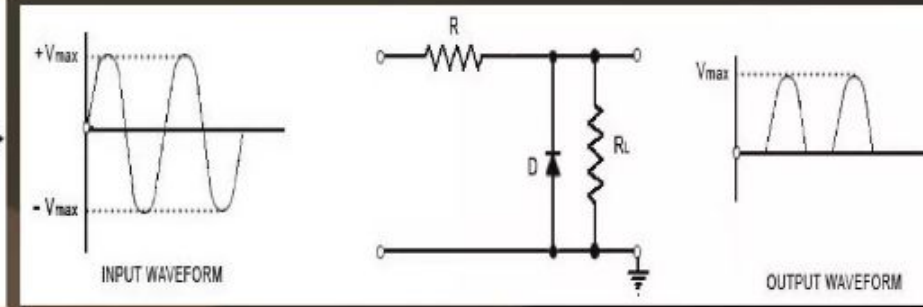
The negative clipping circuit is almost same as the positive clipping circuit, with only one difference. If the diode in figures (a) and (b) is reconnected with reversed polarity, the circuits will become for a negative series clipper and negative shunt clipper respectively.

Negative Series Clipper and Negative Shunt Clipper





(a)



(b)

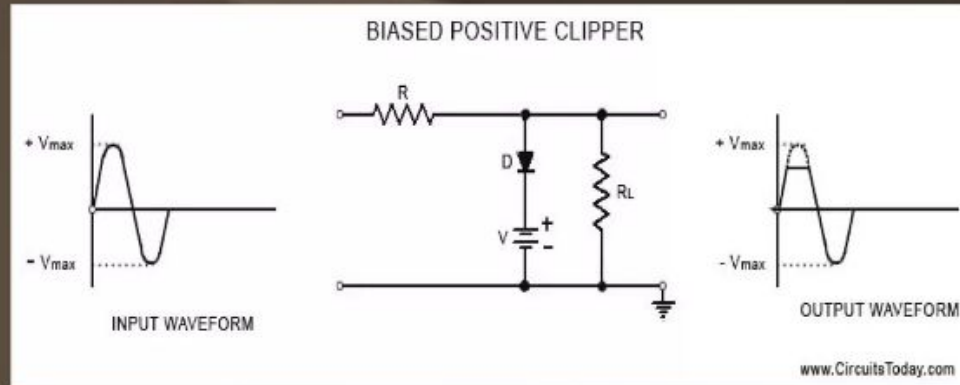
In Figure (a), Anode is connected to the power supply and the cathode is maintained at ground potential.

- During Positive Half Cycle: Output voltage (V_O) = ($V_{in} - V_d$) Volts
- During Negative Half Cycle: Output voltage (V_O) = 0 Volts

In Figure (b), Cathode is connected to the power supply through a resistor R and anode is maintained at ground potential.

- During Positive Half Cycle: Output voltage (V_O) = V_{in} Volts
- During Negative Half Cycle: Output voltage (V_O) = $-V_d$ Volts

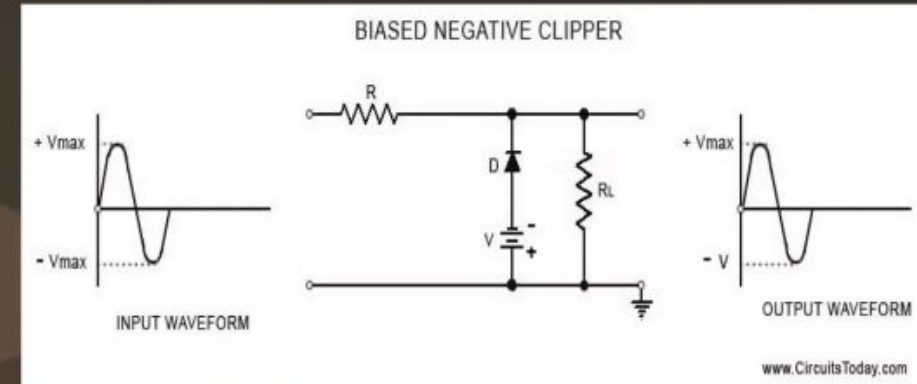
Biased Positive Clipper And Biased Negative Clipper



(a)

A biased clipper comes in handy when a small portion of positive or negative half cycles of the signal voltage is to be removed. When a small portion of the negative half cycle is to be removed, it is called a biased negative clipper.

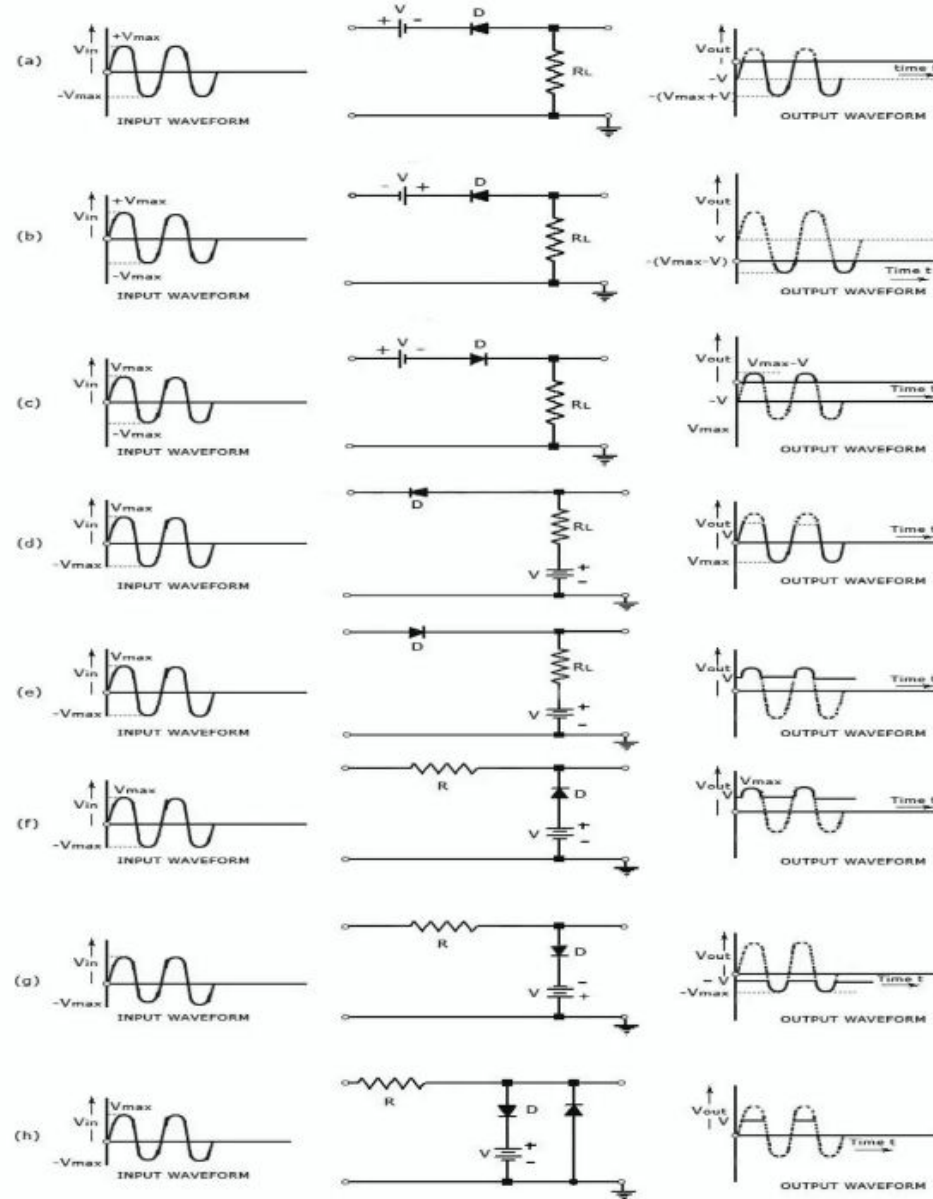
In a biased clipper, when the input signal voltage is positive, the diode 'D' is reverse-biased. This causes it to act as an open-switch. Thus the entire positive half cycle appears across the load, as illustrated by output wave-form [figure (a)]. When the input signal voltage is negative but does not exceed the battery voltage 'V', the diode 'D' remains reverse-biased and most of the input voltage appears across the output. When during the negative half cycle of input signal, the signal voltage becomes more than the battery voltage V, the diode D is forward biased and so conducts heavily. The output voltage is equal to '- V' and stays at '- V' as long as the magnitude of the input signal voltage is greater than the magnitude of the battery voltage, 'V'. Thus a biased negative clipper removes input voltage when the input signal voltage becomes greater than the battery voltage. Clipping can be changed by reversing the battery and diode connections, as illustrated in [figure (b)].



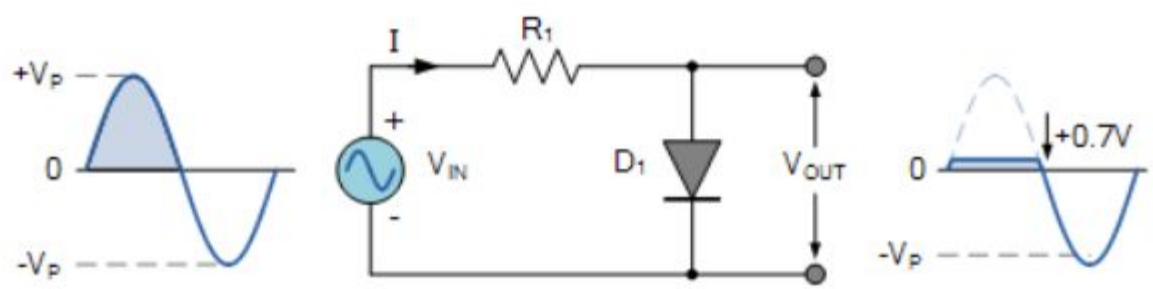
(b)

Different Clipping Circuits

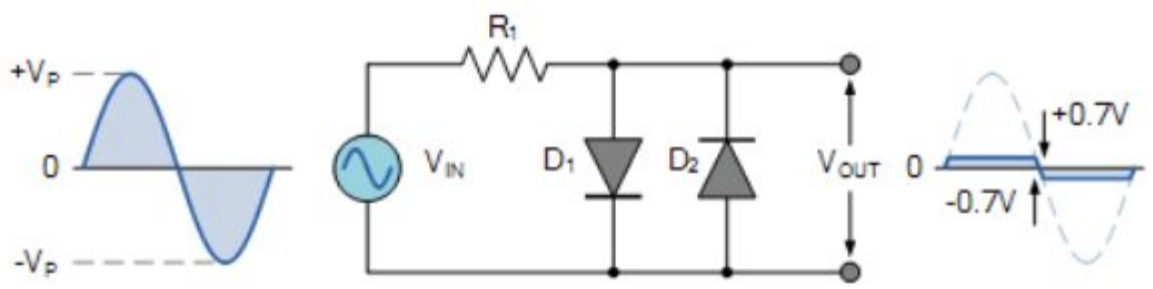
Different Clipping Circuits



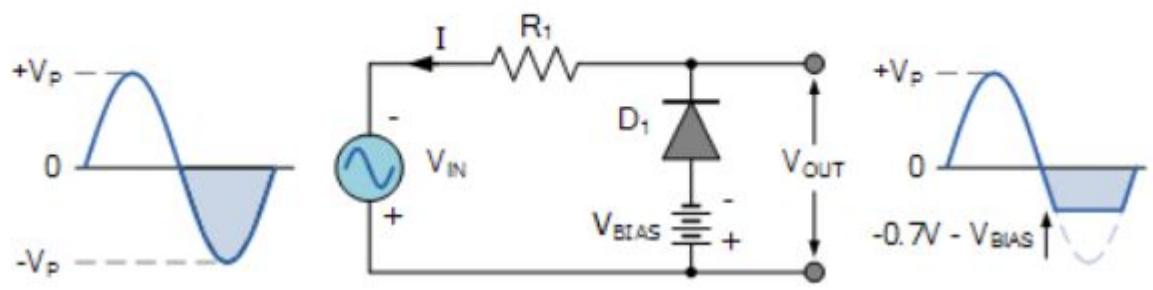
Positive Diode Clipping Circuits



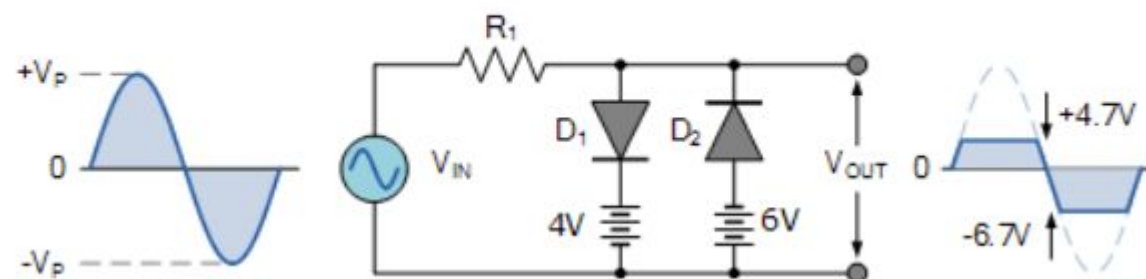
Clipping of Both Half Cycles



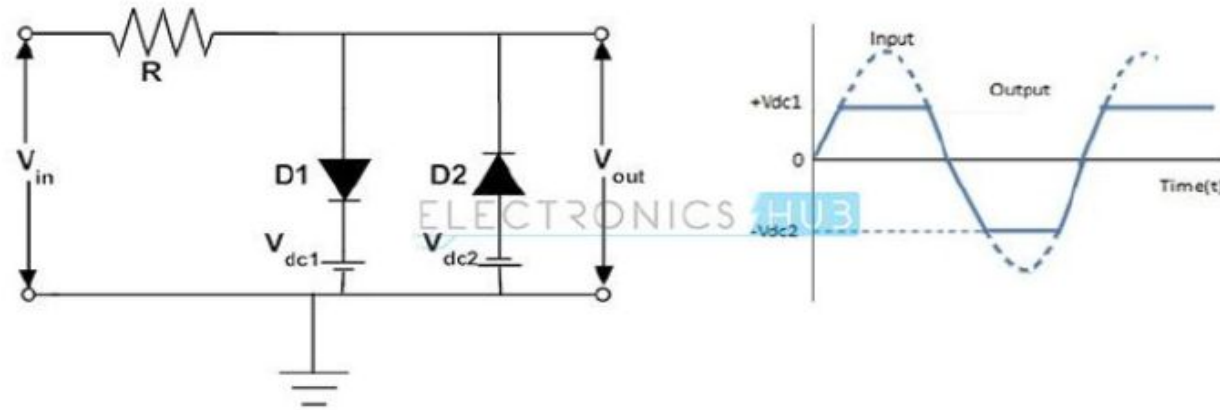
Negative Bias Diode



Diode Clipping of Different Bias levels



Circuit Problem 8. Determine the output voltage and input-output waveform of the figure below. Assuming silicon diodes, 1 kilo-ohms resistor, input voltage of $20 V_P$, and bias voltages of $10 V$.



@ Positive Half-Cycle ($V_{in} = 20V$)
 D_1 is forward-biased.

$$20V - I(1k) - 0.7 - 10V = 0$$

$$I = 9.3 \text{ mA}$$

$$20V - (1k)(9.3 \text{ mA}) - V_o = 0$$

$$V_o = \mathbf{10.7 V}$$

@ Negative Half-Cycle ($V_{in} = -20V$)
 D_2 is forward-biased

$$-20V - I(1k) + 0.7V + 10V = 0$$

$$I = -9.3 \text{ mA}$$

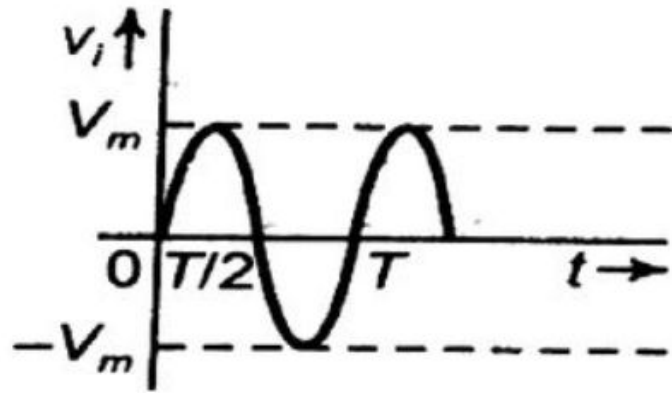
$$-20V - (-9.3 \text{ mA})(1k) - V_o = 0$$

$$V_o = \mathbf{-10.7 V}$$

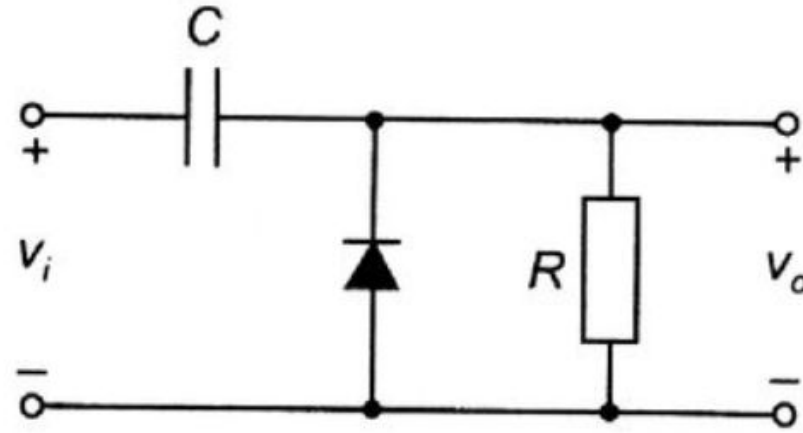
Clampers (Electronic Circuits)

- Also called **DC Restorers**.
- It clamps (or holds, or ties) either the positive or the negative peak of a signal to a definite level.
- The circuit has a **capacitor**, a **diode**, and a **resistor**.
- In addition, it may have a dc supply to introduce additional shift.
- Time constant $\tau = RC$ is made much larger than T (time period) of the signal.
- The capacitor does not discharge when diode is not conducting.

Positive Clamper

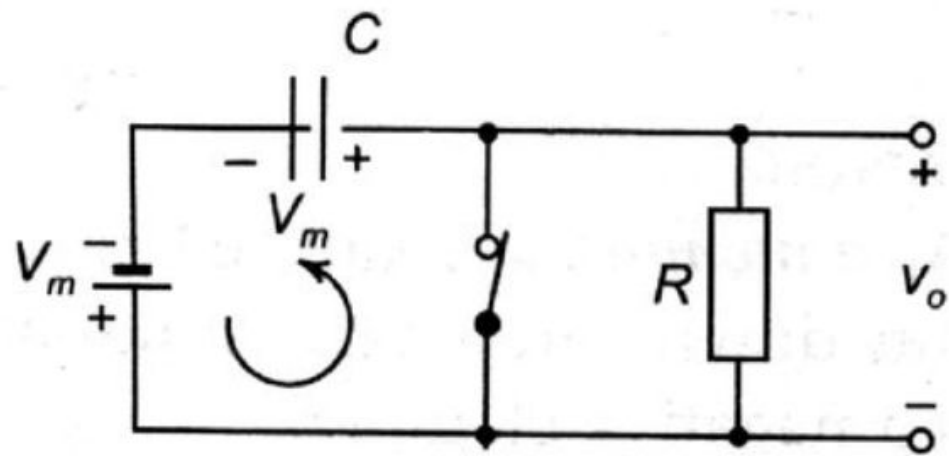


(a) The input.

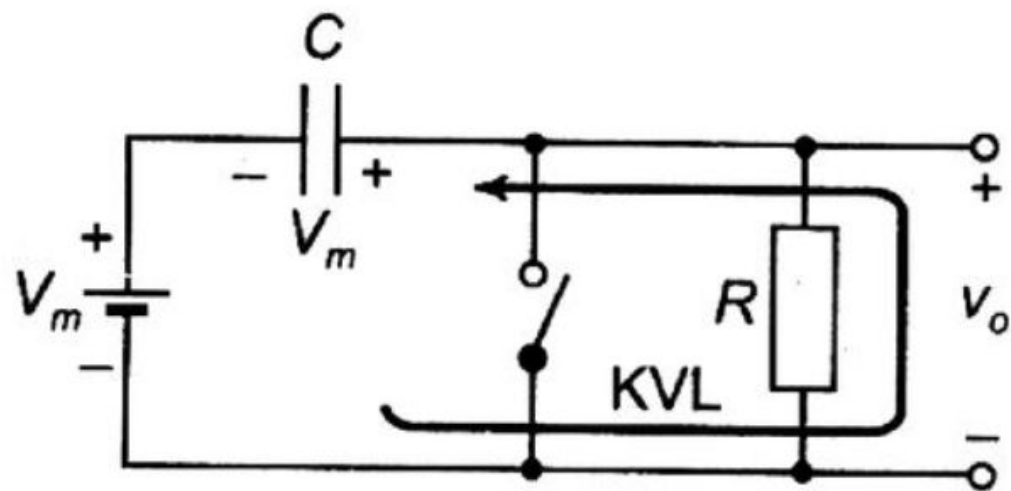


(b) The circuit.

- On first negative cycle, the diode turns ON. The capacitor starts charging. At negative peak, the circuit is as shown in (c). The capacitor charges to V_m .

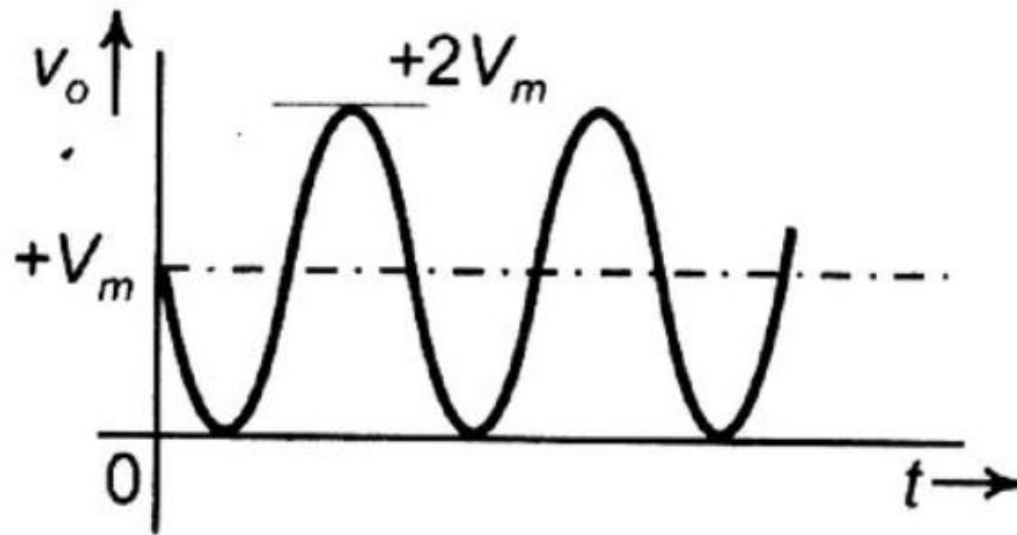


(c)



(d)

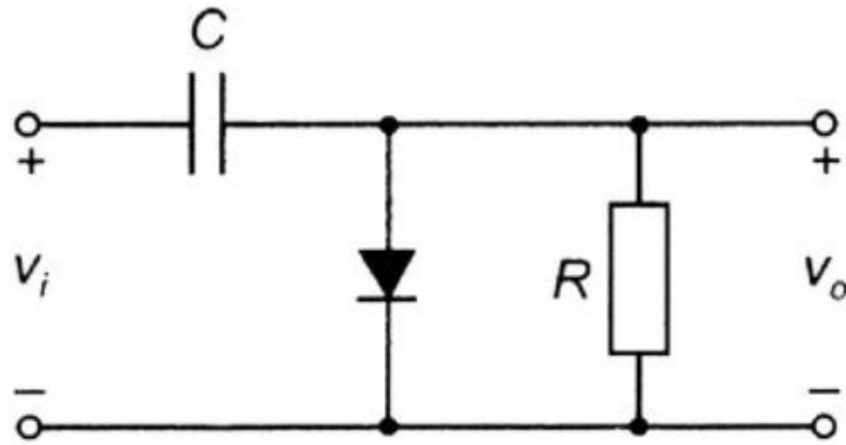
Slightly beyond negative peak, the diode turns OFF. Capacitor does not discharge much because of high RC . At positive peak, the circuit is as shown in (d). The net output is shown in (e).



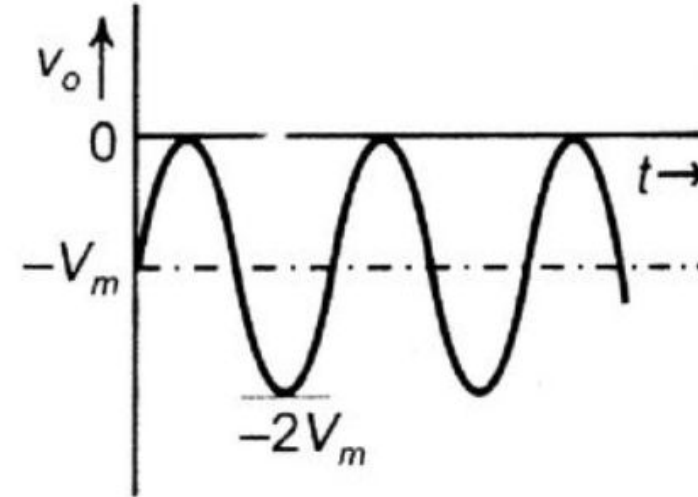
(e) The output.

The charged capacitor acts like a battery of V_m . This is the dc voltage that is added to the signal, as seen in (e). The output sits on 0 V level. The output is **shifted in positive direction.**

Negative Clamper



(a) The circuit.



(b) The output.

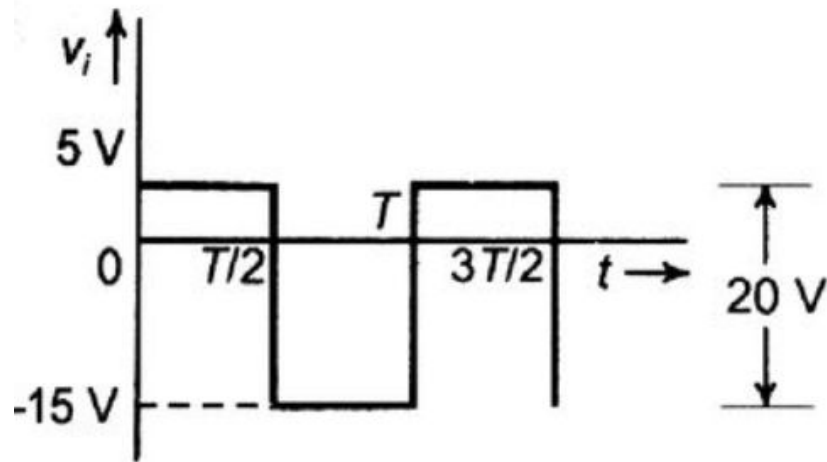
- The diode is turned around. The capacitor voltage reverses, and the circuit becomes negative clamper. It is clamped to zero-volt level, but always remaining below 0 V.
- **Memory Aid** : The diode points in the direction of shift.

Note :

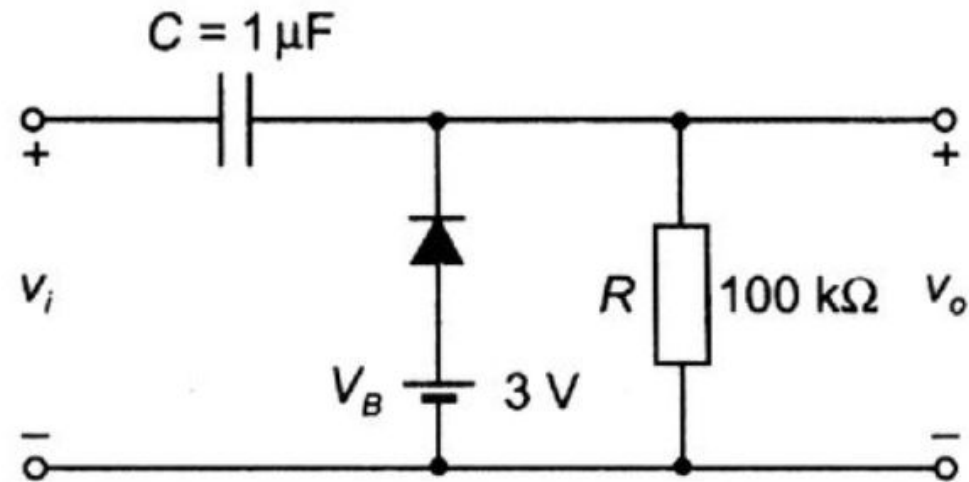
- The total swing of the output is the same as that of the input.
- A clamper can also have an added dc voltage. It is then called **biased clamper**.
- Start the analysis of the circuit for that part of input, for which diode is ON.
- Assume that the capacitor charges to voltage level decided by the circuit.
- Assume that when diode is OFF, the capacitor does not discharge.

Example 1

Determine the output for the input of 1 kHz shown.

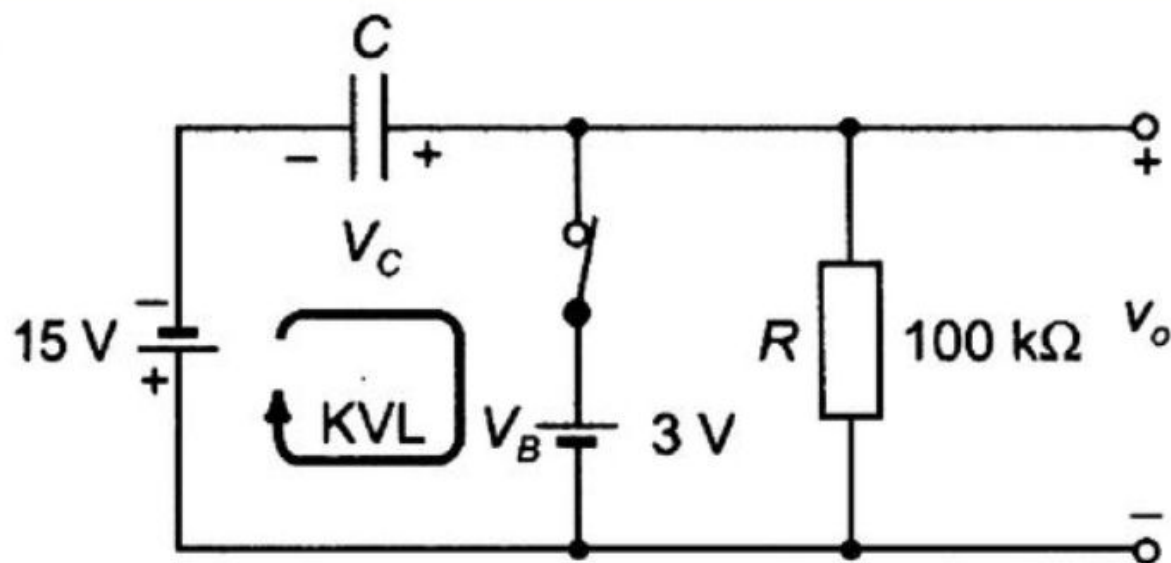


(a) The input.



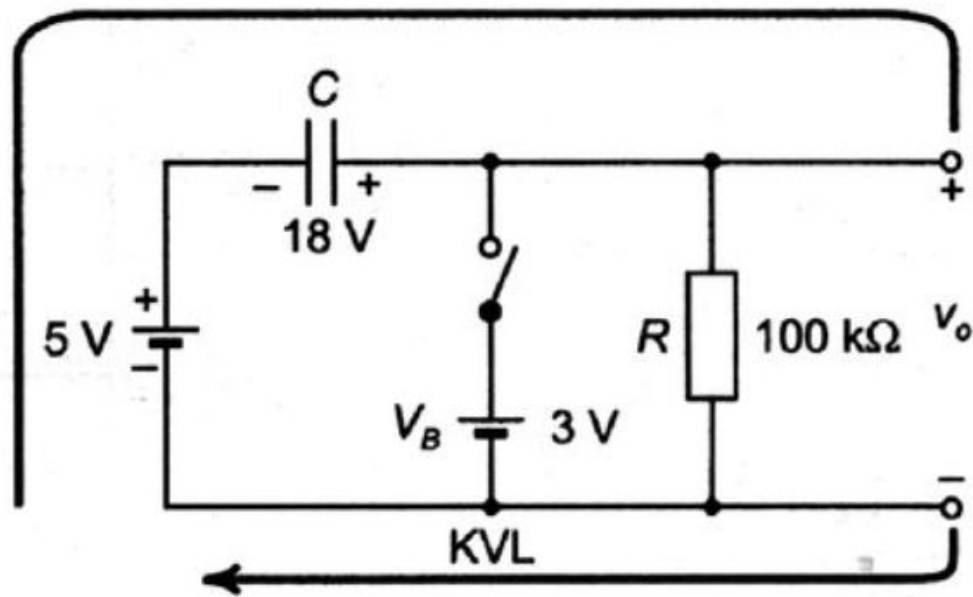
(b) The circuit.

- Is it a positive or negative clamper ?
- Ans. : Biased positive clamper.

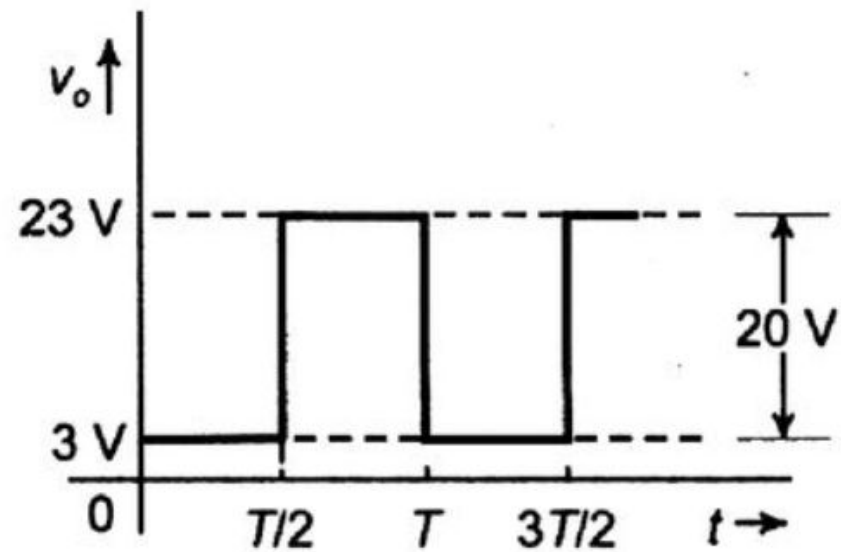


(a)

- The output is across R , but it is also across 3-V battery.
- Hence, $v_o = 3\text{ V}$, during this period.
- Applying KVL, $-15 + V_C - 3 = 0$; $V_C = 18\text{ V}$.
- For the period from T to $3T/2$, the circuit is as shown in (b).



(b)



(c)

- Applying KVL, $v_o = 5 + 18 = 23$ V.
- Thus, the output is as shown in (c).
- **Note** that the output swing is also 20 V.

Clamping networks

Clamping Networks

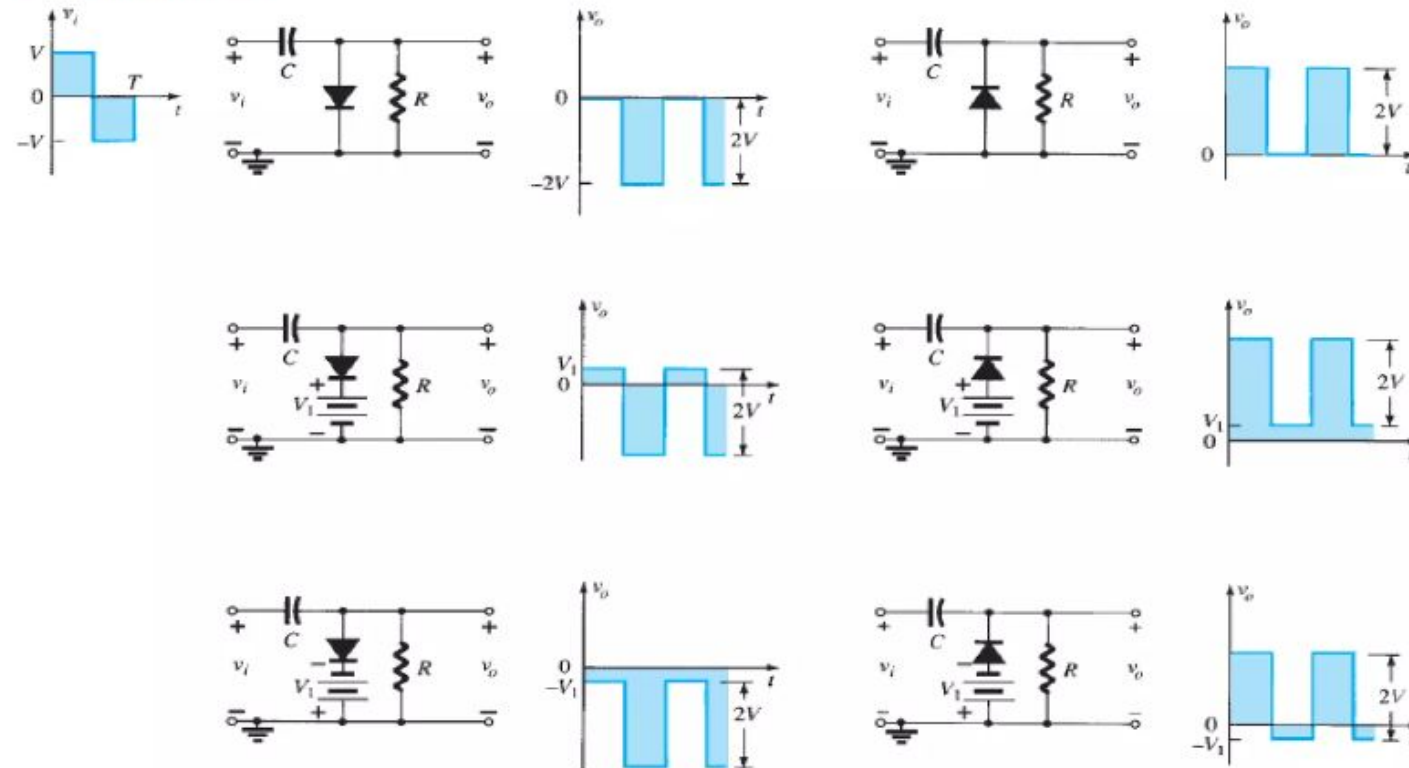
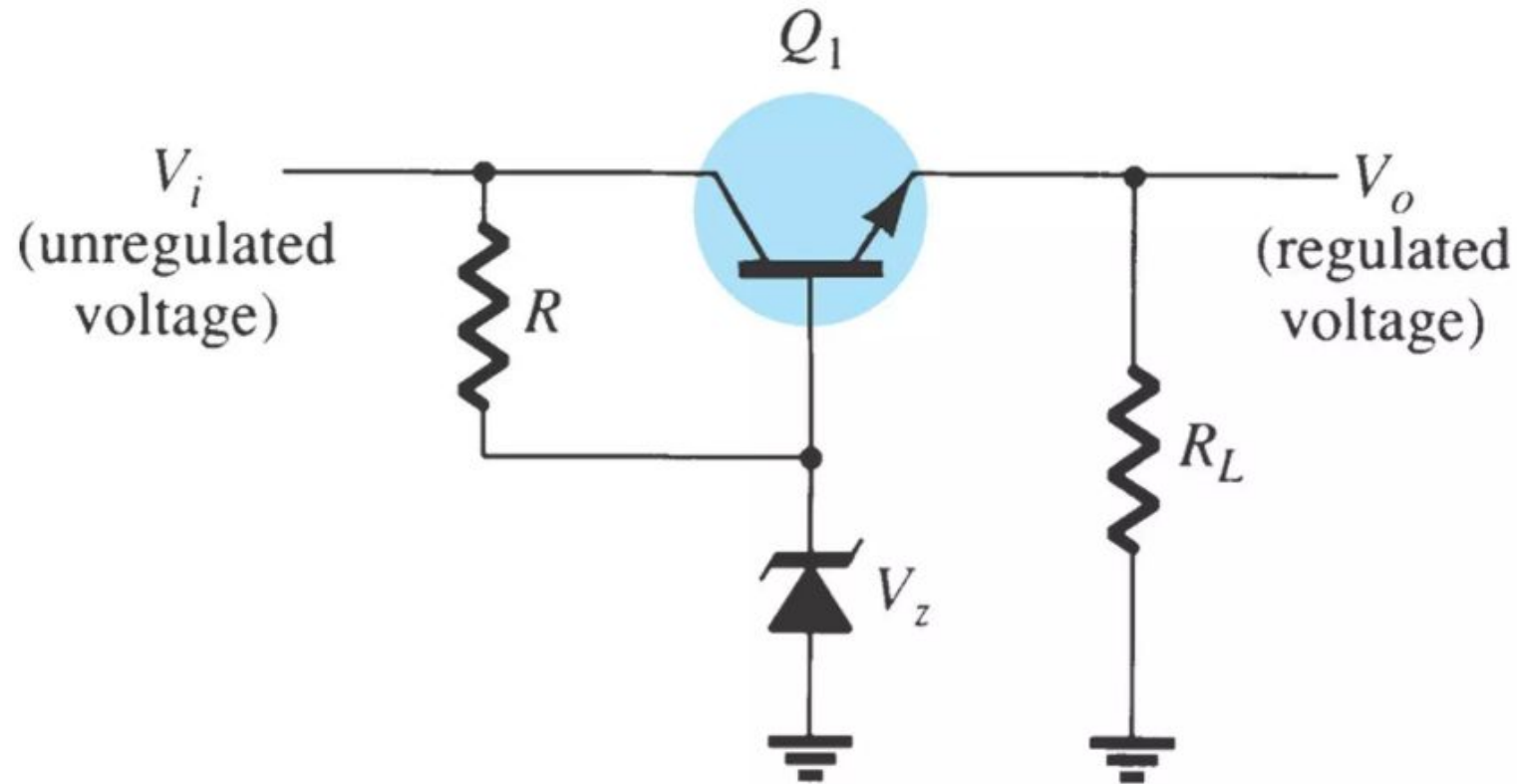


FIG. 2.100

Clamping circuits with ideal diodes ($5\tau = 5RC \gg T/2$).

Transistor Series Regulator



- The transistor Q_1 is the series control element.
- Zener diode provides the reference voltage.

Transistor Series Regulator

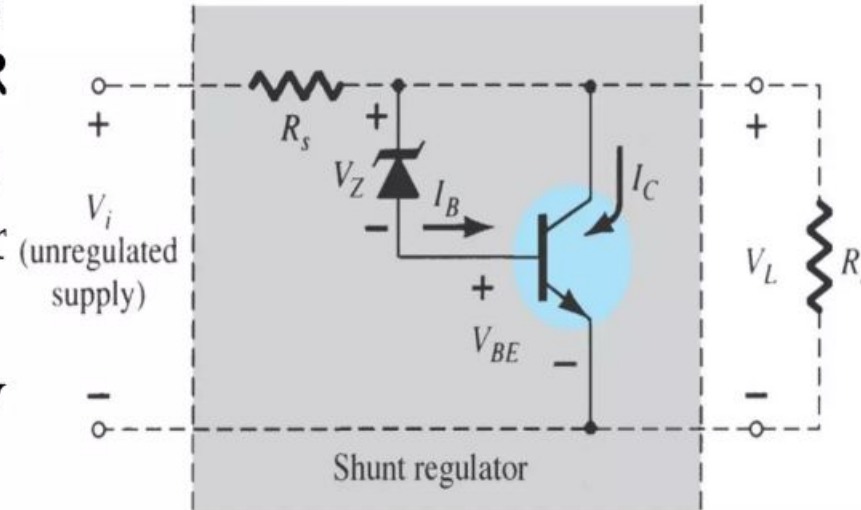
- Since Q_1 is an npn transistor, V_o is found as:

$$V_{BE} = V_Z - V_o$$

- the response of the pass-transistor to a change in load resistance as follows:
 - If load resistance increases, load voltage also increases.
 - Since the Zener voltage is constant, the increase in V_o causes V_{BE} to decrease.
 - The decrease in V_{BE} reduces conduction through the pass- transistor, so load current decreases.
 - This offsets the increase in load resistance, and a relatively constant load voltage is maintained
-

Transistor Shunt Regulator

- Resistor R_s drops the unregulated voltage depends on current supplied to load R
- Voltage across the load is set by zener diode and transistor base-emitter voltage.
- If R_L decrease, a reduced drive current to base of $Q_1 \rightarrow$ shunting less collector current.
- Load current, I_L is larger, maintaining the regulated voltage across load.

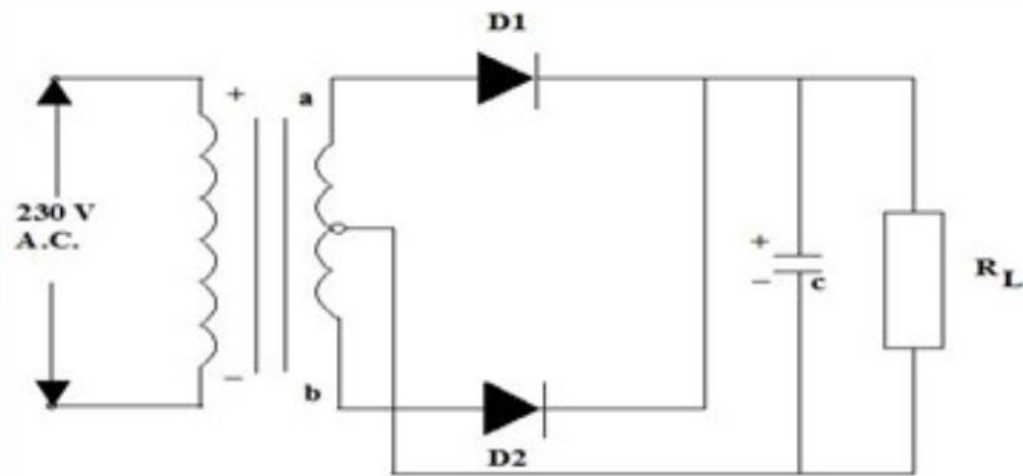


- The output voltage to the load is: $V_o = V_L = V_Z + V_{BE}$
- voltage across the load is set by the Zener diode voltage and the transistor base-emitter voltage.
- If the load resistance decreases, the load current will be larger at a value of: $I_L = \frac{V_L}{R_L}$
- The increase in load current causes the collector current shunted by the transistor is to be less: $I_C = I_S - I_L$
- The current through R_s : $I_S = \frac{V_i - V_L}{R_S}$

WHAT IS A FILTER CIRCUIT?

- ELECTRONIC FILTER CIRCUITS ARE THOSE WHICH PERFORM SIGNAL PROCESSING FUNCTIONS ESPECIALLY TO ELIMINATE UNWANTED SIGNAL, TO ENHANCE WANTED SIGNAL OR BOTH.
- IN FILTER CIRCUITS WE WILL UNDERSTAND MAINLY ABOUT ACTIVE FILTERS AND PASSIVE FILTERS.

Shunt capacitor filter

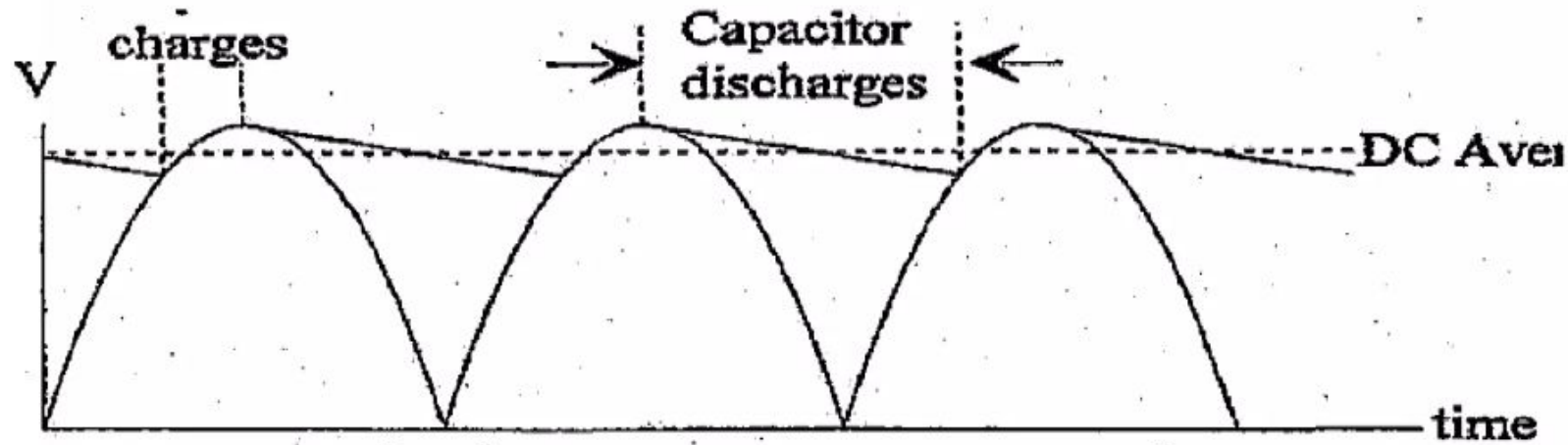
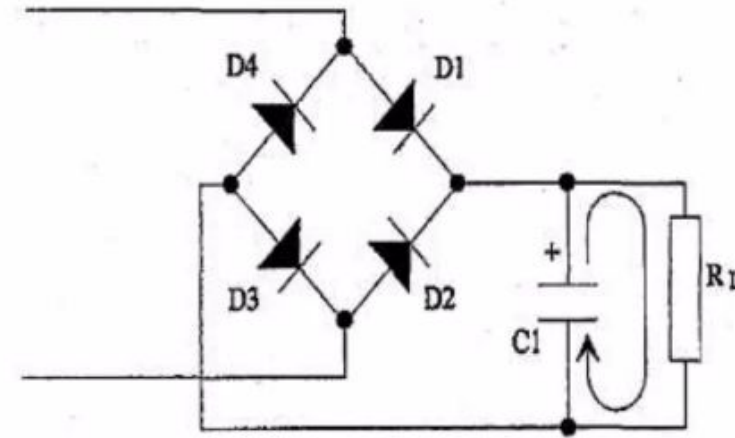
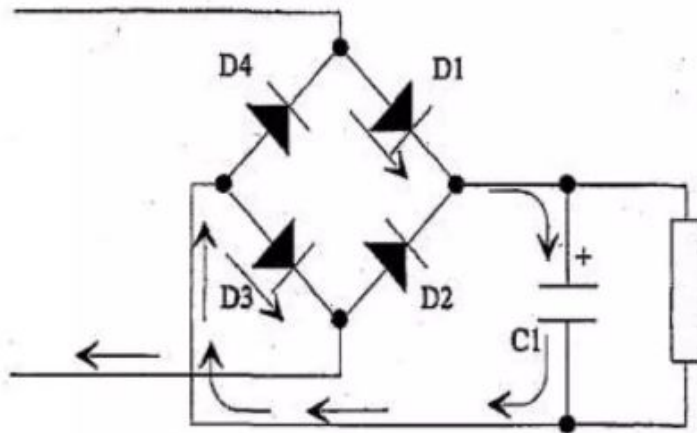


SHUNT CAPACITER FILTER

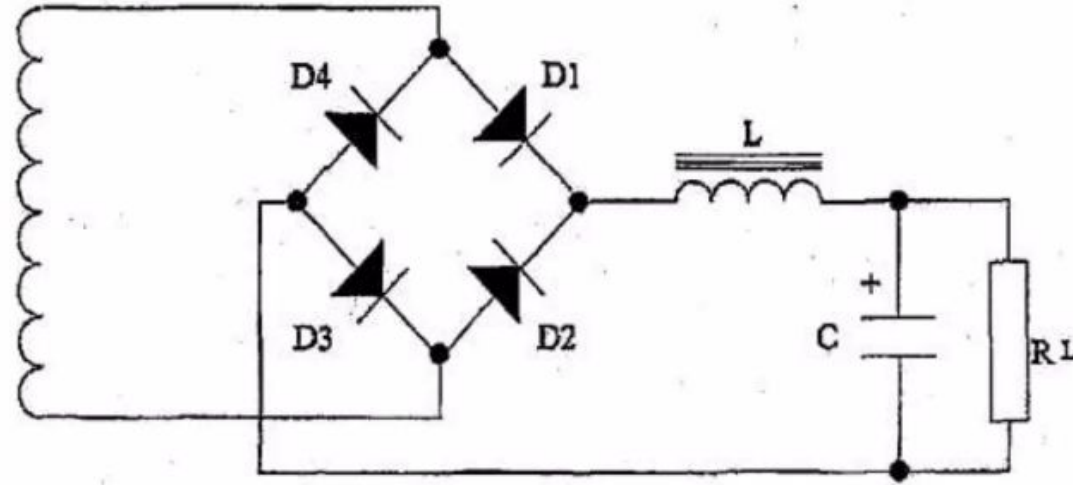


D.C. OUTPUT VOLTAGE WAVEFORM

Filtering with a Capacitor



LC or L Section Filter



L section filter connected to a full wave bridge rectifier

The inductor will operate as an inductor filter, attempting to keep the current flowing through the load at a constant rate. The reduced ripple voltage from the inductor will then be filtered by the action of the capacitor as illustrated

Example 5.1. A sinusoidal voltage of 40 V and frequency 50 Hz is applied to a half-wave rectifier. $R_L = 200\ \Omega$, $V_F = 0$, $R_F = 20\ \Omega$, $R_r = \infty$. Find V_{dc} , I_{dc} , I_{max} , I_{rms} , P_{dc} , η , ripple factor, % regulation.

[M.D. Univ. Analog Electronics, May-]

Solution:

$$V_{S\ rms} = 40\ V$$

$$\text{Frequency, } f = 50\ \text{Hz, } R_L = 200\ \Omega, R_F = 20\ \Omega, R_r = \infty$$

$$I_{max} = \frac{V_{S\ max}}{R_L + R_F} = \frac{40\sqrt{2}}{200 + 20} = 0.257\ \text{A or } 257\ \text{mA Ans.}$$

$$I_{rms} = \frac{I_{max}}{2} = \frac{257}{2} = 128.5\ \text{mA Ans.}$$

$$I_{dc} = \frac{I_{max}}{\pi} = \frac{257}{\pi} = 81.8\ \text{mA Ans.}$$

$$V_{dc} = I_{dc} R_L = 81.8 \times 10^{-3} \times 200 = 16.36\ \text{V Ans.}$$

$$P_{dc} = I_{dc}^2 R_L = (81.8 \times 10^{-3})^2 \times 200 = 1.338\ \text{W Ans.}$$

$$P_{ac} = \frac{I_{max}^2}{4} (R_F + R_L) = \frac{(257 \times 10^{-3})^2}{4} \times (20 + 200) = 3.63\ \text{W}$$

$$\eta = \frac{P_{dc}}{P_{ac}} \times 100 = \frac{1.338}{3.63} \times 100 = 36.86\% \text{ Ans.}$$

$$\text{Ripple factor, } \gamma = \left(\frac{I_{rms}^2}{I_{dc}^2} - 1 \right)^{1/2} = \sqrt{\left(\frac{128.5}{81.8} \right)^2 - 1} = 1.21 \text{ Ans.}$$

$$V_{dc} = \frac{V_{max}}{\pi} - I_{dc} R_F = \frac{40\sqrt{2}}{\pi} - 81.8 \times 10^{-3} \times 20 = 18 - 1.636 = 16.364\ \text{V}$$

$$\begin{aligned} \text{Regulation} &= \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100 = \frac{\frac{V_{max}}{\pi} - \left(\frac{V_{max}}{\pi} - I_{dc} R_F \right)}{\frac{V_{max}}{\pi} - I_{dc} R_F} \times 100 = \frac{18 - 16.364}{16.364} \times 100 \\ &= 9.998\% \text{ Ans.} \end{aligned}$$

